



Busy Airports: Predicting TSA Traffic at Major U.S. Airports

The Erdős Institute
Data Science Boot Camp Project
Spring 2026

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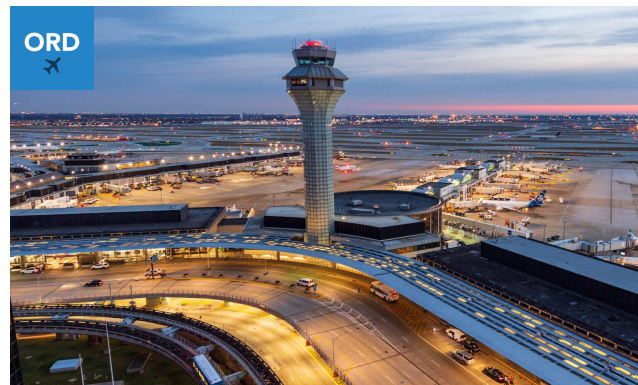
Introduction



Goal: Develop a predictive model for passenger throughput at TSA security checkpoints for airports in the US using Chicago O'Hare International Airport (ORD) as our case study.

Motivation:

- Effective Staffing and Resource Allocation
- Airport Operational Decision-Making
- Improving the Traveler Experience



Question: Can we accurately predict passenger throughput at TSA security at ORD?

Data Gathering and Dataset Creation



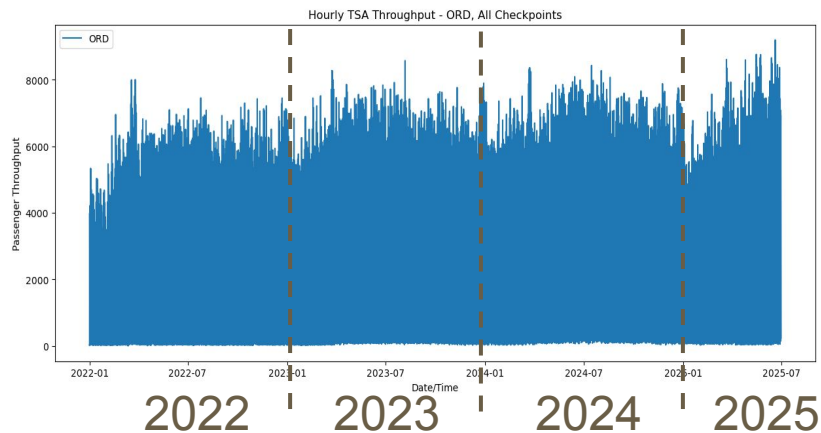
- Origin Data: [TSA Weekly Passenger Throughput Data](#)
- Data Format: Raw data is stored in weekly PDFs for the years 2022-2025
- Dataset Creation:
 - Built custom python scripts ([scripts/](#) directory) to extract the values from the PDFs
 - Create CSV files used to train models
- Project Data:
 - Github: [data/tsa-throughput-data-complete.csv](#)
 - Used methods in [src/busy_airports/preprocessing.py](#) to clean up column names



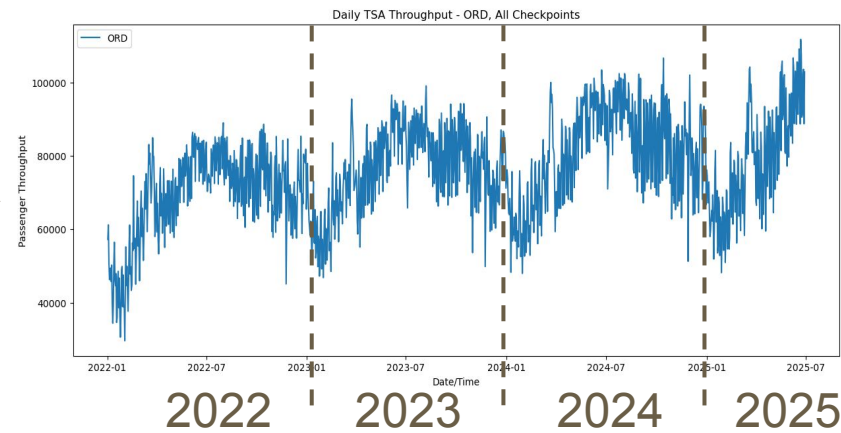
Exploratory Data Analysis



To enhance forecasting accuracy, we aggregate hourly data into daily data, as the former exhibits excessive noise and volatility.



Hourly Data

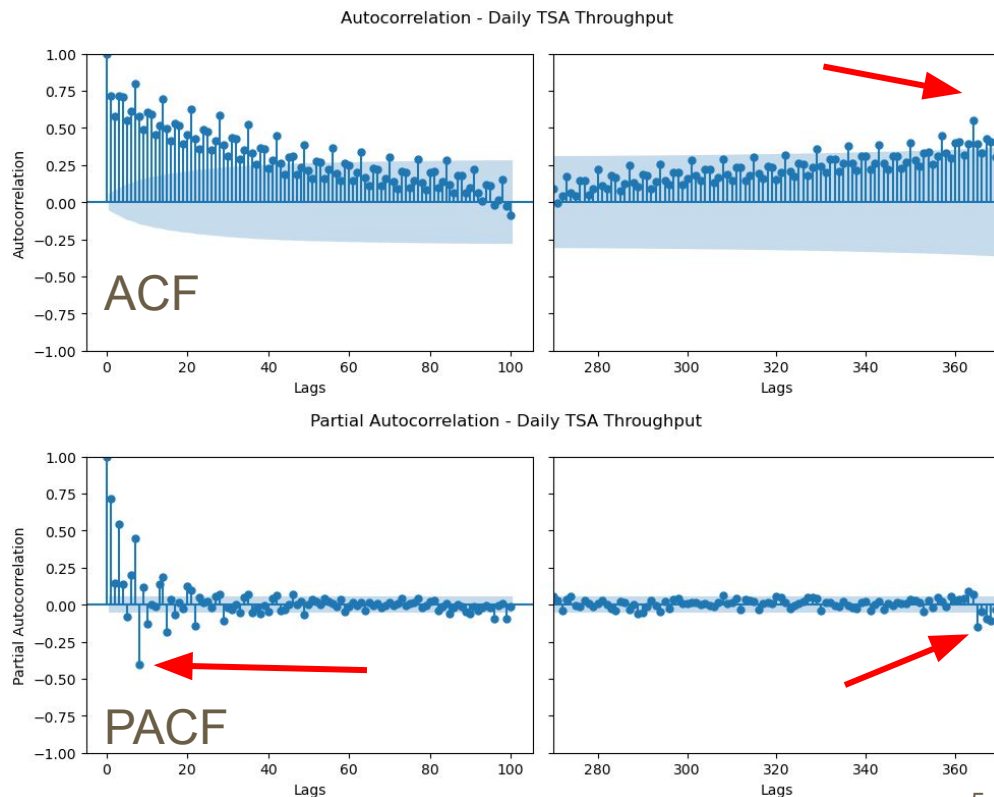


Daily Data

Exploratory Data Analysis: Seasonality

Analysis of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots from the complete dataset revealed significant spikes at lags **7** and **365**.

- This suggests the presence of **weekly** and **yearly** seasonality, respectively.

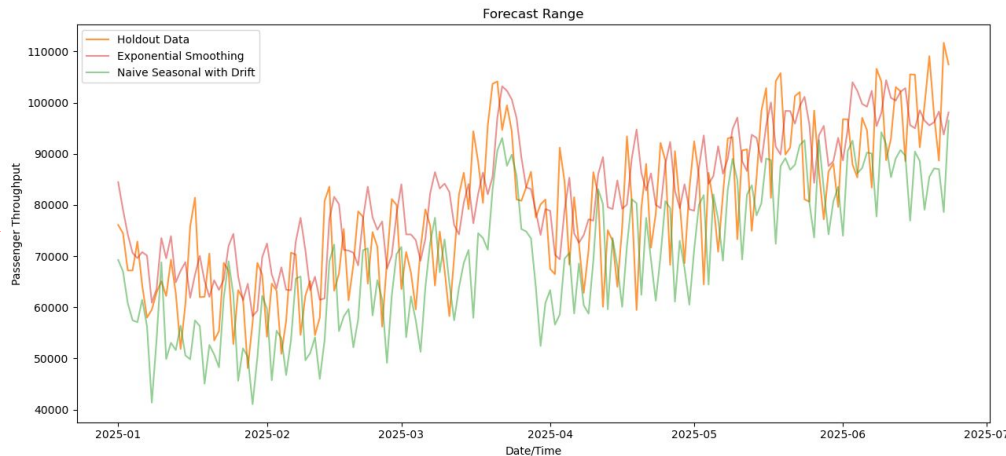
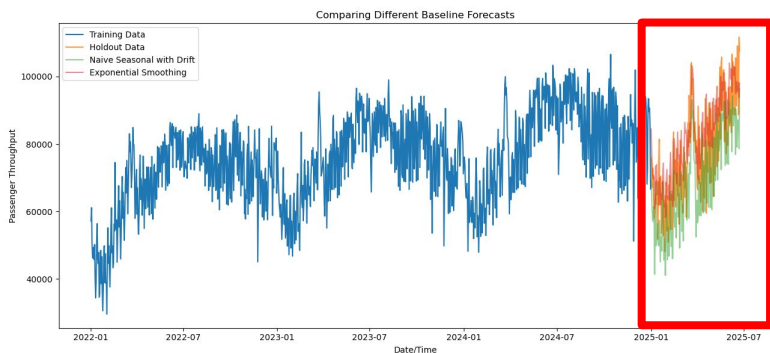


Baseline Models



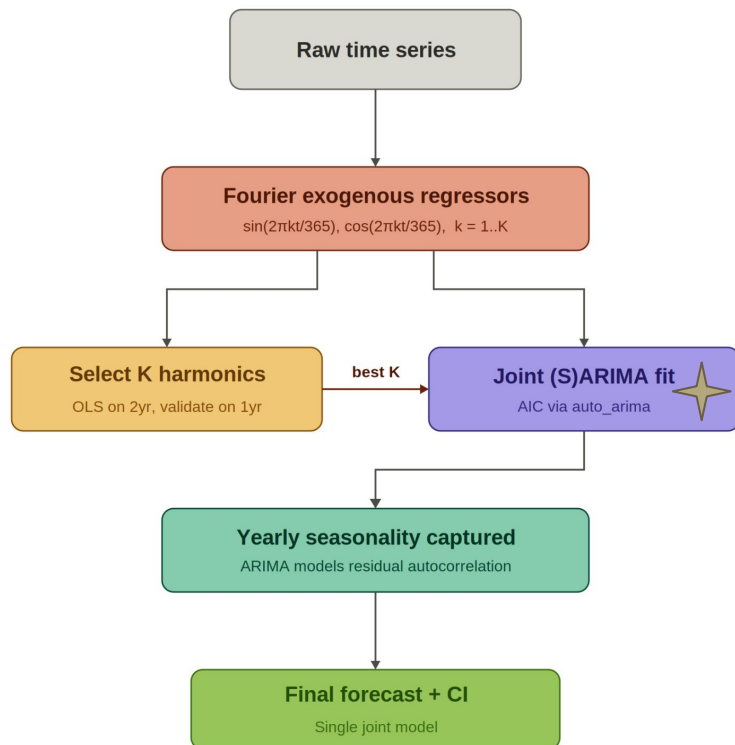
- **Naive Seasonal Model with Drift (NSWD):** Accounts for repetitive seasonal patterns and overall growth/decline.
- **Triple Exponential Smoothing (ExpS):** Captures level, trend, and seasonality.

These are simple enough models that are able to capture the **annual seasonality** and **overall trend** that we see in the passenger data.

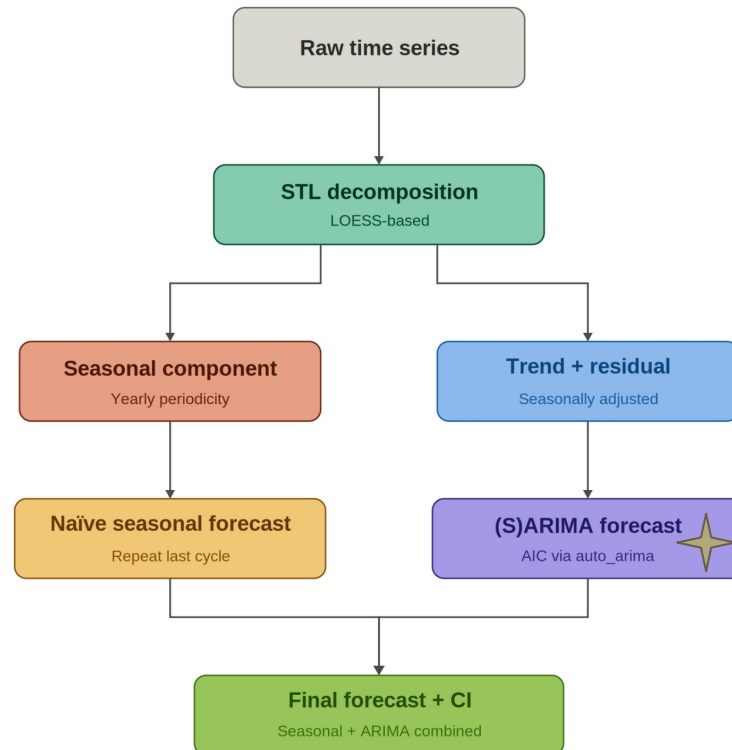


Modeling Approach

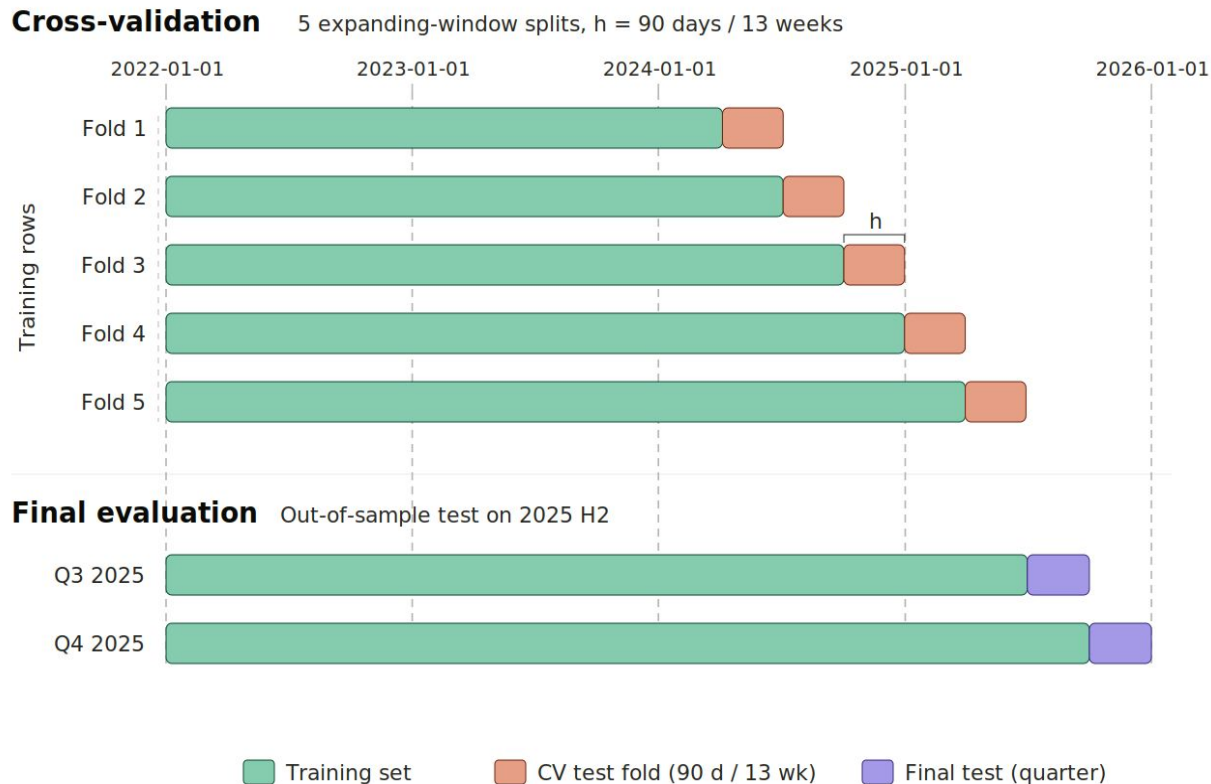
Harmonic regression + (S)ARIMA



Seasonal-Trend decomposition with LOESS (STL) + (S)ARIMA

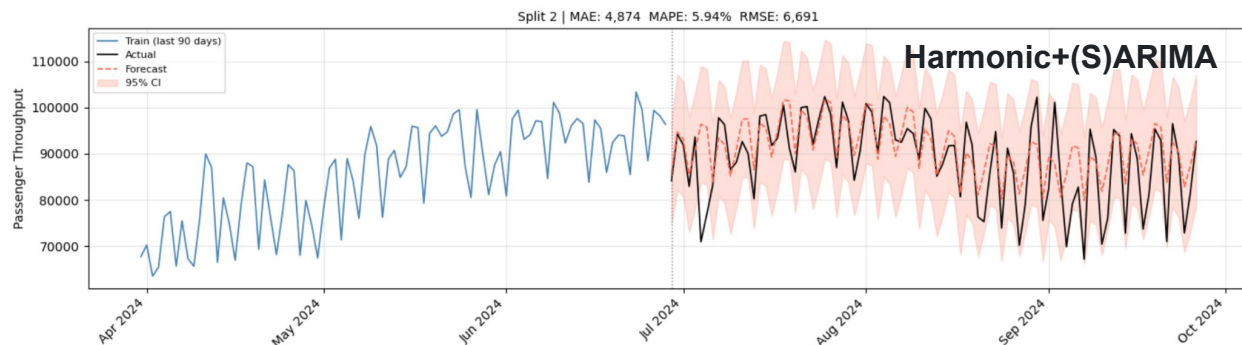
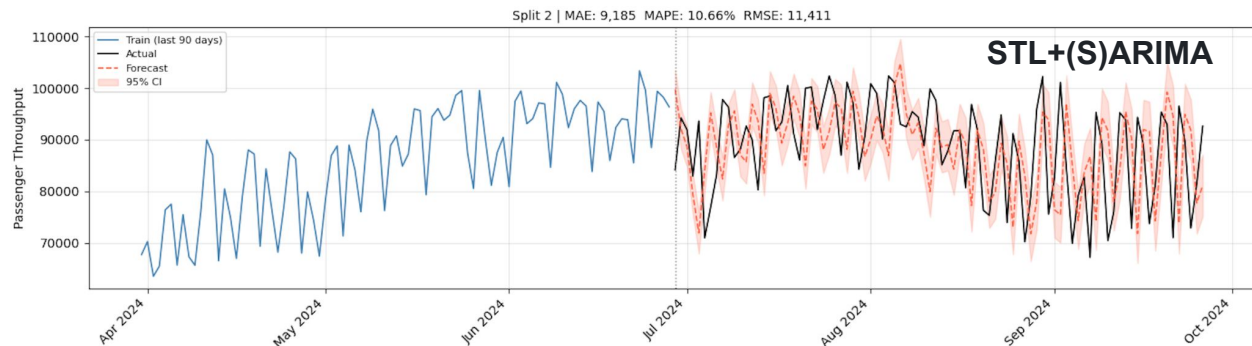


Time Series Splits (Cross-Validation for all models)



Results: Validations

Cross-validation results indicate that Harmonic Regression + (S)ARIMA is the better performing model here due to its significantly lower MAPE.

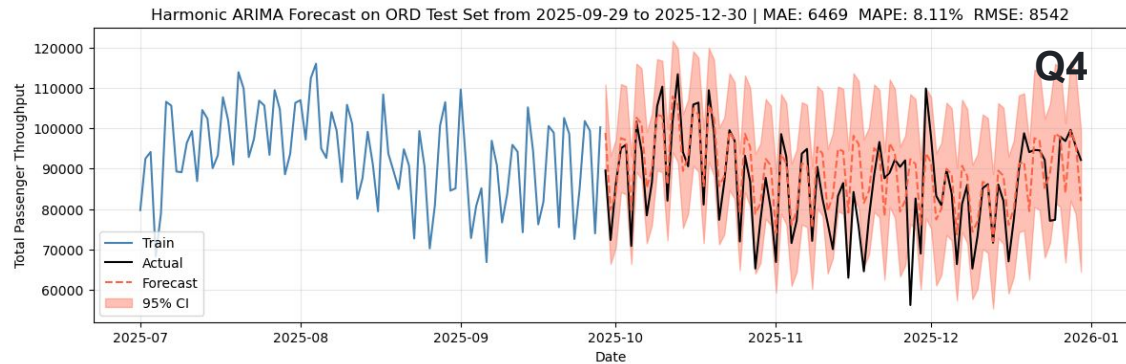
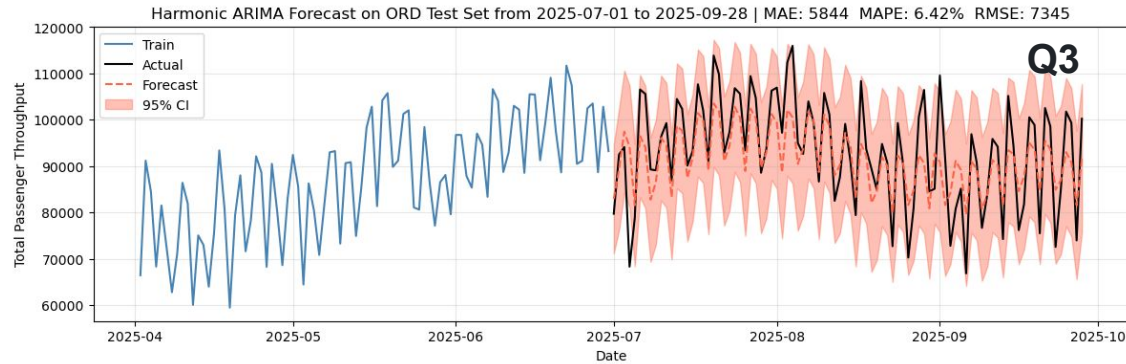


Model	MAE (pass./day)	RMSE (pass./day)	MAPE
NSWD (baseline)	15,263	17,905	18.82%
ExpS (baseline)	9,368	11,434	11.99%
STL+(S)ARIMA	9,190	11,430	11.65%
Harmonic+(S)ARIMA	6,432	7,984	7.83%

Results:

HR + (S)Arima

Final evaluation on Q3 and Q4 2025 test data showed a result of **6.42%** and **8.11%** MAPE, respectively.

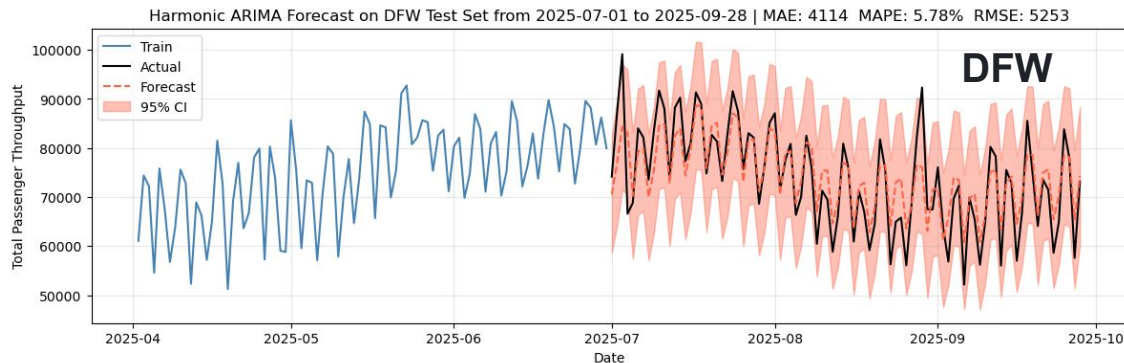
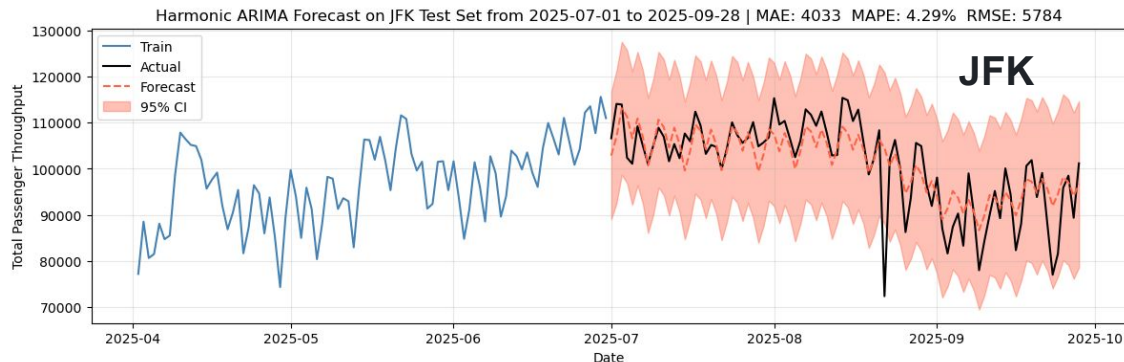


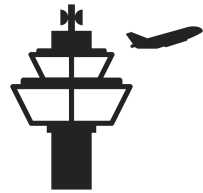
Time Period	MAE (pass./day)	RMSE (pass./day)	MAPE
2025-Jul to 2025-Sep	5844	7345	6.42%
2025-Sep to 2025-Dec	6469	8542	8.11%

Results:

Other Airports

Airport	MAPE (Q3 2025)	MAPE (Q4 2025)
ORD	6.42%	8.11%
JFK	4.29%	5.69%
DFW	5.78%	9.38%
LAX	4.85%	7.36%





Conclusion and Next Steps

- We trained and cross-validated forecasting models to predict the daily passenger throughput at TSA checkpoints in Chicago O'Hare Airport.
- The **Harmonic Regression + (S)ARIMA** model proved to be the most effective, beating baseline models and predicting the total passenger throughput to within less than **10% mean average percentage error**.
- The model also extrapolated well to other similar airports, demonstrating robustness and ability to generalize to unseen airport data.
- **Future Work:** Apply other models (XGBoost, Prophet, etc.). Extend this model to capture different types of airports (small airports, different geographic regions) using a combined time series/regression approach.